

Fluid Flow in Pipes

Pre-Lab Plan
Unit Operations Lab 4
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Group 1, Tuesday, 1 PM Section
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1. Experimental Objective

The objective of the lab *Fluid Flow in Pipes* is to utilize experimental data such as flow rates and pressure drops to investigate key sections of the fluid-flow system. The main sections under examination include rotameter, orifice meter, various pipe types such as a U-Tube, Elbow, Expansion and contraction valve. The Reynolds number is used to determine various parameters such as pipe friction head losses, pipe fitting head losses and the fanning friction factor.

2. Experimental

2.1 Apparatus

The main system under investigation is the pipe system. The pump serves as the means of allowing water flow into the system. Various pipe fittings are pictured which are to be investigated,



Figure 1 Main Apparatus

1) Reservoir Tank	Used to ensure constant water flow to pump.
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 Figure 2 Tank	
2) Reservoir release valve  Figure 3 Pump Valve	Valve to allow water to exit reservoir tank
3) Pump power switch and pump  Figure 4 Pump	Switch to turn on power and pump is utilized to circulate water throughout the system,
4) Pump inlet valve  Figure 5 Pump Bypass Valve	Used to allow water to enter pump

5) Flow rate meter <i>Figure 6 Flow rate meter</i>	Reads flowrate of air in SCFM must be converted to LPM using correction factor
6) Rotameter <i>Figure 7 Rotameter'</i>	Reads pressure drop across wherever the tubes are connected.
7) Water flow control valves <i>Figure 8 Water flow valves</i>	Controls flow rate of water into the system.

8)  Figure 9 Tubes of various diameter	Tubes of various diameter to be investigated
9) Various pipe fittings  Figure 10 Various Pipe fittings	Pipe fittings to be investigated in lab (SU-Tube, Expansion, Contraction, Elbow)
9)  Figure 11 Pipe fittings 10)	Pipe fittings used to connect to rotameter to measure pressure drop.

2.2 Materials and Supplies

Type	Material/Supply	Purpose
Material	Water	The main fluid that goes through the pipe system and is used to measure the pressure drop and analyze various characteristics of the flow (velocity and more)
Supply	Reservoir Tank/Buckets	To hold the water supply and collect the fluid when calibrated through the rotameter, measuring the volume in time.

Supply	Graduated Cylinders	To measure the water's volume collected over time (5s for pre-lab) to obtain the volumetric flow.
Supply	Stopwatch/Timer	To accurately measure the time collecting a specific volume of water when calibrating the rotameter.
Supply	Plastic Tubes with Disconnect Fits	This was to connect the pressure taps on the pipes to the pressure gauge.

2.3 Experimental Plan (10 pts) -

Pre-Lab Preparation

1. Read and understand the lab manual before entering the laboratory.
2. Wear proper attire and safety equipment. Upon entering, put on the required PPE (lab coat, gloves, safety goggles, etc.).

Day 1 – Determination of conversion factor (SCFM to LPM)

Objective:

Establish a conversion factor between air flow rate (SCFM) and corresponding water flow rate (LPM) using the pump and flow measurement system.

Procedure:

1. Ensure the water tank outlet valve is closed. Open the water hose and begin filling the water tank [1]. (*[#]- references an equipment labeled above)
2. Continue filling until the water level is above the inlet pipe feeding into the pump but below the pump discharge pipe. This prevents air entrapment in the pump.
3. If the pump fails to deliver flow upon startup, release trapped air by opening the vent valve located on the discharge pipe.
4. Prime the pump if necessary [3]. Consult the instructor if difficulties occur.
5. Activate the pump and verify continuous waterflow through the system.
6. Confirm that all downstream valves after the flow meter are open prior to taking measurements.
7. Adjust the waterflow rate using the control valves located to the left of the flow meter [7].
8. Measure water flow by collecting the discharged water in a graduated container for a measured time interval.

9. Record the following data for each trial:
 - a. Flow rate in SCFM (from water flow meter [5])
 - b. Collection time (s)
 - c. Collected water volume (mL)
10. Repeat measurements for multiple water flow settings to obtain a calibration range.
11. Calculate the conversion factor from SCFM to LPM based on measured data.
12. The experimentally determined correction factor was 0.89. This might vary per group.
13. Drain the water tank upon completion using the tank release valve [2] and ensure the system is dry.

Day 2 – Pressure Drop Across Pipe Geometries

Objective:

Measure the pressure drop across various pipe geometries at different waterflow rates and determine the relationship between geometry and pressure loss.

(*geometries will refer to any fittings, bends, etc)

Procedure:

1. Identify all pipe geometries to be tested (elbows, bends, expansions, contractions, valves, etc.) [8,9].
2. Connect the pressure drop measuring device to the geometry of interest using two flexible hoses equipped with penetration fittings [10].
3. Attach hoses to the designated inlet and outlet ports of the geometry. Ensure correct orientation according to labeled connections on the device. Reversed connections will result in zero readings.
4. Verify a secure connection of the fittings. Slight water leakage from one hose may occur, so place a waste container beneath the connection to prevent spillage.
5. Initiate system operation at the desired waterflow rate and allow the flow to stabilize before recording data.
6. Measure and record the pressure drop across the geometry using the rotameter [6].
7. Conduct at least two to three replicate measurements for each geometry to ensure reproducibility and precision.
8. Repeat the procedure for all geometries under three waterflow conditions (e.g., 3, 6, and 9 SCFM).
9. Record all measurements in a structured data sheet.
10. Once finished drain the water tank and ensure the system is as dry as possible
11. Check with instructor before leaving the lab.

Experimental Variables:

- **Independent Variables:** waterflow rate (SCFM to LPM) and pipe geometry type.
- **Dependent Variables:** Pressure drop (ΔP) across geometry, friction factor, Reynolds number

Project Safety Evaluation (10 pts)

	<i>Yes/No</i>	
Potential electrical hazards?	Yes	If yes, identify conditions: Relating to electrical parts like the pump, the meters for reading the differential pressure transmitters. The system is also a bench that is with a reservoir which allows for a shock hazard related to spills or closeness to water.
Potential mechanical hazards?	Yes	If yes, identify conditions: Related to the pump as there are moving parts within it. The water plugs/fittings need to be properly used to prevent leaks and other failures. The control valves to bring a potential mechanical hazard.
Potential pressure hazards?		If yes, identify conditions: Also related to the pump and containing pressurized water. Too much pressure could result in a system failure and water sprays that are difficult to control.
Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	Yes	If yes, specify type: Regular standard nitrile gloves should be worn for lab safety and cleanliness.
MSDS reviewed by all team members?		List MSDS sheets reviewed: Water (CAS# 7732-18-5) has been reviewed. All materials in the safety have been reviewed as well.
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	Lab room temperature (Ambient)	Location: Fluid Flow pipe system Hazard: None Precaution: Watch for the thermocouple as it has the temperature necessary for the density and viscosity calculations.

Pressure (psia)	Determined from the pump	Location: Pump (flow path) Hazard: Water leaks or burst could occur if the pressure of the pump is too high or the fittings are loose. Precaution: Try to operate the pump at a safe spot and make sure the fittings are tight and secure before starting the pump
Safety Controls	yes/no	<i>If yes, Provide description</i>
System Flow Procedure	Yes	<i>Team must close the valve to restrict the flow of water before pulling out and connecting the fittings. The valve can be opened only after connecting the tubes to the next section.</i>
DPT Calibration/Zeroing	Yes	The DPTs need to be checked for a correct zero reading (0-0.02) each time before use. If the zero-reading on the DPT is wrong, contact the TA for help in re-zeroing.
DPT Flushing	Yes	The DPT needs to be flushed through opening and closing the black valves on its side after connecting to the new pressure taps. This is to clear the air pockets.
Electrical Awareness	Yes	The pump and other electrical components need to be kept dry and team must not touch electrical switches with wet hands.

I have read relevant background material for the Unit Operations Laboratory entitled: Fluid Flow in Pipes and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Ify Bidokwu

signature of team member

10/15/2025

date

Miguel Piza

signature of team member

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Christian Salinas Rojas

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